to produce fruit, animals to eat these fruits, or even when and how a cell ought to divide. All these entities, without a central authority, use a set of rules to operate independently and produce an outcome that produces goodness within the system. This is often a natural way of designing systems that can scale to millions or billions of participants. Blockchains, similarly, compel us to reconsider some, not all, centralised structures in our human-made world.

Interestingly, we are already beneficiaries of human-made distributed systems. The internet, for example, is one of humankind's best built distributed system that has scaled to what it is today precisely because it is distributed: there is no central regulator for how the internet expands or what devices you can use to connect to it. In a similar vein, computer scientists, have, since the 1960s, developed a slew of algorithms and systems for distributed ledgers. They have found use in systems for stock trading, air traffic control, or even file sharing. Hence, in some sense, blockchains are not entirely new. They are part of a continuum of scalable distributed system technology that the computing industry has been building by using nature as inspiration.

While the internet may be a distributed system, most of the use cases built atop it—banking, social media, gaming—are, in fact, centralised and run by companies such as HDFC, Meta and Epic. These centralised use cases, often referred to as Web 2.0, result in concentrated power in the hands of a few companies. This is where blockchain technology offers a disruption to the status quo, by allowing the opportunity to rebuild such use cases (or new ones altogether) that are decentralised, effectively meaning that there is no central entity or owner. Such use cases are referred to as Web 3.0.

Web 3.0 enthusiasts believe that reimagining end-user applications in decentralised fashion will reduce or remove the corporate intermediaries that can otherwise produce higher costs, censorship and opacity, and other types of friction. In fact, crypto founders and users often referenced these arguments to support the idea of a universal distributed, digital currency separate from the monetary policies of central banks. While crypto has suffered major setbacks in 2022, other applications of blockchain are being actively explored that hold great promise for India and the world in 2023 (and beyond). These include:

- Tokenisation of real-world assets, such as houses, so that owners could list their property on a blockchain and sell equity in it, and such transactional details would be stored on the ledger;
- India lacks legacy digital systems to track, for example, property/land records. This is a valuable use case to track such ownership, streamline individual creditworthiness, and ensure smoother credit flow to millions of self-employed individuals.
- As a deterrent for fraud in banking by creating a single-point KYC, distributed verification and validation;
- Entire new paradigms for identity management for individuals seeking jobs. This replaces traditional background checking mechanisms in place today. Blockchains can enable end users to control who they expose their information to, when, and why.
- Health records systems that can consolidate information from across centralised systems into a single blockchain to provide doctors/patients with a complete view of health;
- Platforms for self-publishing of music or other types of content that can have baked-in royalty fees so that creators can earn each time their content is used.

INDIA LACKS LEGACY DIGITAL SYSTEMS TO TRACK PUBLIC RECORDS. THIS CAN BE AN **OPPORTUNITY FOR NOVEL APPLICATIONS** THAT LEVERAGE BLOCKCHAINS TO BUILD SCALABLE PUBLIC INFRA

India, like many other countries, lacks legacy digital systems to track health, property, and other forms of public records. But this lacuna is an opportunity for a novel set of applications that leverage blockchains to build scalable public infrastructure. Much like in the early days of the internet, when Netscape 1.0 was the most exciting invention, it seemed the killer use case of the web was a bookmarking service, which is what Yahoo! used to be. But clearly, today the internet is so much more than that. Similarly, the current era of the blockchain is in its infancy and we do not yet know what the perfect use case will be. But the hunt is certainly on.

I am an eternal optimist about technology. And although blockchain technology might not have delivered the "stickiest" use cases yet, we should not be in a hurry to write off this technology because of recent setbacks related to one application: cryptocurrency. Yes, governance and regulations in this space need to improve. But entrepreneurs—especially those here in India—should not stop seeking those game-changing use cases of blockchains in the meantime. ■